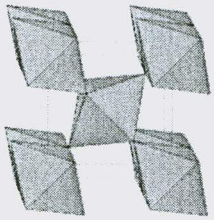
CUME

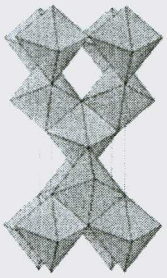
Inorganic Chemistry

Nov-12-2010

On the past Monday, Dr. Michel Dupuis in his seminar discussed two TiO_2 crystal structures:



Rutile



Anatase

Facts:

a). Anatase is active and rutile is inactive in photocatalysis.

b). TiO_2 is an important inorganic material in drug pill, white paint, bone implant, sunscreen, etc.

Questions:

(1) What are the basic building block(s) in anatase, and that in rutile? (10')

(2) Indicate which crystal structure(s) involving corner-sharing, and which involving edge-sharing. (10')

(3) Predict which crystal structure is more thermal-stable than the other, and explain why so. (10')

(4) Explain why TiO_2 can be used in drug pill, bone implant, and sunscreen. (10')

(5) Which crystal structure could be better than the other in sunscreen? List your reason(s). (10')

(6) Which could be an insulator, and which a wide band-gap semiconductor? Give your reasons. (10')